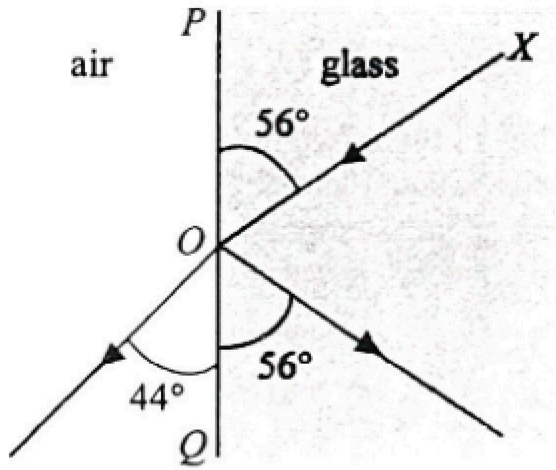
ch18 Refraction of Light

Items:

2018 Q17

17.



In the figure above, XO is a light ray incident on the glass-air bou
following gives the refractive index of glass ?

A. $\frac{\sin 56^\circ}{\sin 44^\circ}$

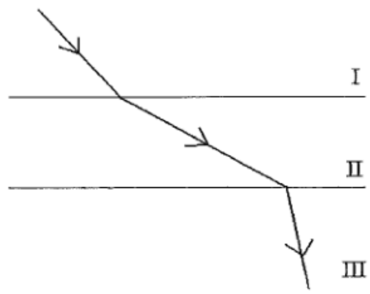
B. $\frac{\sin 44^\circ}{\sin 34^\circ}$

C. $\frac{\sin 56^\circ}{\sin 46^\circ}$

D. $\frac{\sin 46^\circ}{\sin 34^\circ}$

2014 Q15

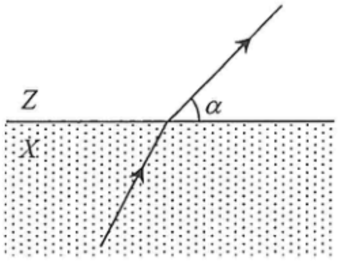
15. The figure shows the path of a light ray travelling from medium I to medium III separated by parallel boundaries. Arrange in **ascending order** the speed of light in the respective media.

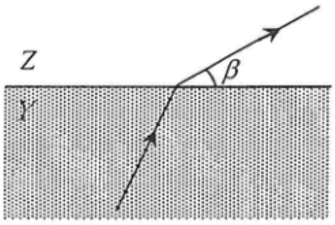


A. I < III < II
B. II < III < I
C. III < I < II
D. III < II < I

2020 Q13

13. Monochromatic light travels with the same incident angle from media X and Y respectively to another medium Z as shown.



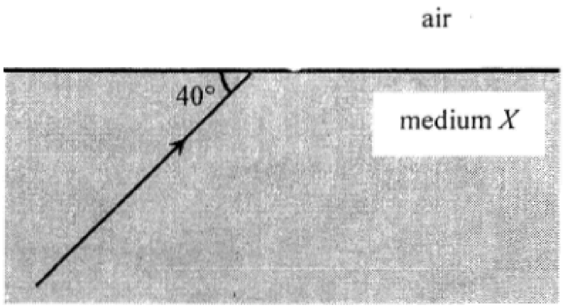


The corresponding refracted rays in Z make angles α and β respectively with the boundary plane (with $\alpha > \beta$). Which medium, X or Y, has a greater refractive index ? In which medium, X or Y, does light travel faster ?

	medium with a greater refractive index	medium in which light travels faster
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

2013 Q20

20.



A ray of light is travelling from a transparent medium X to air making an angle of 40° with the boundary plane as shown. If the angle between the refracted ray in air and the reflected ray in medium X is 70°, find the refractive index of medium X.

A. $\dfrac{\sin 40°}{\sin 30°}$

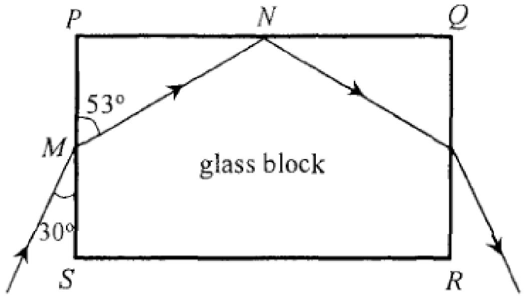
B. $\dfrac{\sin 30°}{\sin 40°}$

C. $\dfrac{\sin 60°}{\sin 50°}$

D. $\dfrac{\sin 50°}{\sin 60°}$

2021 Q13

13.

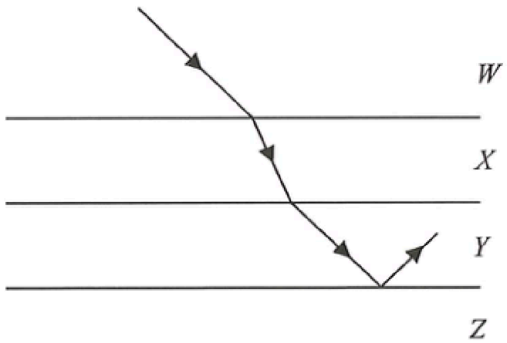


The figure shows the cross-section of a rectangular glass block *PQRS*. A light ray is incident from air at *M* on face *PS* and the refracted ray strikes face *PQ* at *N*. Find the critical angle for the glass-air interface.

A. 37°
B. 44°
C. 53°
D. 60°

2023 Q16

16. The figure shows a light ray passing through the horizontal boundaries between media *W*, *X*, *Y* and *Z*. The light ray in *W* is parallel to that entered into *Y* from *X*. Total internal reflection occurs at the boundary between *Y* and *Z*.



Which medium has the highest refractive index ?

A. *W*
B. *X*
C. *Y*
D. *Z*

2016 Q20

20. A beam of white light is separated into different colours after entering a glass prism because lights of different colours

A. are diffracted to different extents by the prism.
B. undergo total internal reflection at different angles inside the prism.
C. travel at different speeds in vacuum.
D. travel at different speeds in glass.

2013 Q21

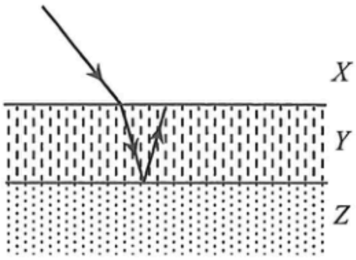
21. White light can be resolved into component colours by using a glass prism. Which of the following statements is/are correct ?

(1) The refractive indices of glass for different component colours are not the same.
(2) Red light travels faster than violet light in a vacuum.
(3) The frequencies of all the component colours are reduced when entering the prism.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2025 Q14

14.

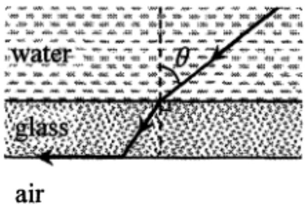


The figure shows three media X, Y and Z separated by parallel boundaries. A light ray travelling from X into Y then undergoes total internal reflection at the boundary between Y and Z. Arrange in descending order the speed of light in the respective media.

A. $X\ Y\ Z$
B. $X\ Z\ Y$
C. $Y\ Z\ X$
D. $Z\ X\ Y$

2016 Q17

17.



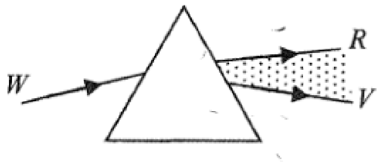
A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle θ on the glass sheet and finally emerges into air along the glass-air interface as shown. Find θ.
Given: refractive index of water is 1.33.

A. 41.2°
B. 48.8°
C. 53.1°
D. It depends on the refractive index of glass.

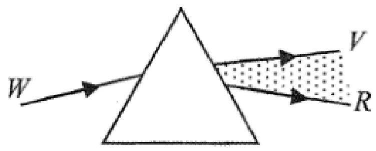
2015 Q17

17. Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively ? It is known that red light travels faster than violet light in glass. (R = red, V = violet, W = white)

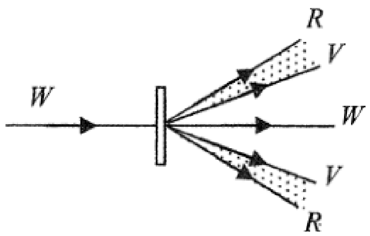
(1)



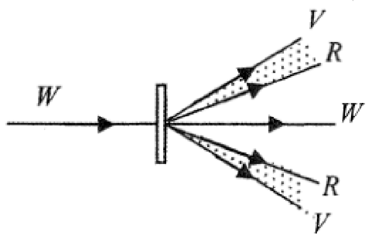
(2)



(3)



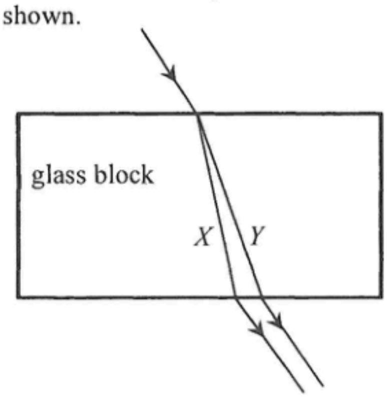
(4)



A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

2024 Q14

14. A light beam consisting of monochromatic light beam X and Y is incident on a rectangular glass block. It splits into two beams as shown.

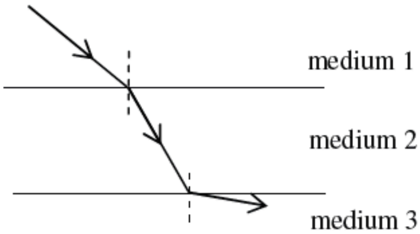


Which of the following comparisons of beam X and Y is/are correct ?

(1) X has a higher frequency.
(2) X has a larger speed in glass.
(3) In the glass block, the critical angle for X is greater.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

20.



As shown above, a ray of light travels from medium 1 to medium 2, and then enters medium 3. The boundaries are parallel to each other. Arrange the speed of light, c, in the three media in **ascending** order.

A. $c_3 < c_2 < c_1$
B. $c_3 < c_1 < c_2$
C. $c_2 < c_3 < c_1$
D. $c_2 < c_1 < c_3$